

2. Users and Stakeholders

2.1 Local User

Role:

A person who uses the system to find a suitable local service provider.

Main goals:

- Submit a household or equipment service problem.
- Select a service type directly when known.
- Get help identifying the service category when unsure.
- Find suitable service providers.
- Track service requests and previous services.

Main tasks:

- Register and log in.
- Enter a problem description.
- Select a service category if known.
- Submit a service request.
- View the suggested service category from the NLP/ML module.
- View matched service providers.
- Send a request to a provider.
- Track request progress.
- View completed services.
- Provide feedback and ratings.

Important characteristics:

- The user may know the exact service needed or may only know the problem.
- The system should use simple language and an easy form.
- The user should not need technical knowledge to describe a problem.

2.2 Service Provider

Role:

A person who provides repair or other local services through the system.

Main goals:

- Receive requests related to their services.
- Decide whether to accept a request.
- Complete requested work.
- Maintain their service information and availability.

Main tasks:

- Register and log in.
- Create a service provider profile.

- Select service categories.
- Update availability.
- View suitable service requests.
- Accept or reject requests.
- Update request status.
- Complete service requests.
- View service history.
- View feedback from Local Users.

Important characteristics:

- A provider should receive requests related to their registered services.
- Providers should only manage their own requests.
- Provider information should be kept accurate and current.

2.3 System Administrator

Role:

Manages the overall system and maintains system records.

Main goals:

- Keep the system organized and usable.
- Manage users and service providers.
- Maintain service categories.
- Monitor service requests.
- Maintain accurate records.

Main tasks:

- Log in to the administrator area.
- Manage Local User accounts.
- Manage Service Provider accounts.
- Approve or deactivate provider accounts.
- Add, update, or remove service categories.
- View service requests.
- Monitor request status.
- Manage inappropriate or invalid records.
- View system reports.

Important characteristics:

- Administrator functions must be protected.
- The administrator has more permissions than other users.
- Administrative actions should not be available to normal users.

2.4 NLP/ML Module

Role:

Analyzes the problem description entered by a Local User and predicts an appropriate service category.

Main tasks:

- Receive the user's problem description.
- Process the text.
- Identify important words and patterns.
- Predict a service category.
- Return the predicted category to the system.
- Support the provider-matching process.

Important characteristics:

- The model provides a suggested category, not a guaranteed technical diagnosis.
- The prediction should be based on the training data.
- The system should allow the user to select the correct category when they already know it.